

Week 4 Evaluation Submission

Evaluation One-Liner

I will measure story task completion, child-safe content, personalization coverage, repetition avoidance, output structure, p95 latency, and estimated cost on my Week 3 Kids Daily Story Agent using a LangSmith golden dataset of 43 cases covering happy path, edge cases, known failures, safety, reading level, personalization, history repetition, performance budget, and style scenarios, with code-based evaluators plus trace review. Pass bar: 85% strict case pass rate, 100% structure, 100% child safety, 90% personalization coverage, 95% repetition avoidance, p95 latency under 15 seconds, and estimated story-generation cost under \$0.05 per run. I will run this in LangSmith and report the delta from the baseline run to the post-improvement run.

Framework

Field	Completed Entry
Agent under test	Week 3 Kids Daily Story Agent, a LangGraph parent-reviewed story-generation workflow in <code>src/story_agent.py</code> .
User outcome	A parent gets a safe, age-appropriate, personalized story that is ready for review and can be delivered by email without missing downstream fields. If the story is unsafe, generic, repetitive, malformed, too long, too slow, or too expensive, the agent fails its core job.
Metrics	Quality: strict case pass rate, <code>child_safe_content</code> , <code>required_interests_included</code> , <code>favorite_character_type_included</code> , <code>theme_alignment</code> , <code>moral_alignment</code> , <code>not_repetitive_for_child</code> , <code>output_structure_valid</code> , and <code>age_appropriate_length</code> . Cost and behavior: <code>latency_within_budget</code> , <code>estimated_cost_within_budget</code> , and <code>tool_steps_succeeded</code> .
Judge method	Code-based evaluators score JSON structure, word count, required interests, favorite character type, theme/moral token overlap, unsafe terms, repetition, latency, estimated cost, and tool failures. LangSmith trace review is used for failure examples and trajectory inspection.
Golden dataset	Source is <code>data/eval/golden_story_cases.jsonl</code> , hand-labeled from the agent's expected user workflows and known demo risks. It contains 43 cases across happy path, edge, known failure, safety, reading level, personalization, history repetition, performance, and style categories.
Pass bar	Strict case pass rate $\geq 85\%$; structure 100%; child safety 100%; required interest coverage $\geq 90\%$; favorite character coverage $\geq 95\%$; repetition avoidance $\geq 95\%$; p95 latency $< 15s$; estimated cost $< \$0.05$ per run.
Instrumentation	LangSmith traces end-to-end graph runs, node spans, LLM calls, memory/history/image/audio/PDF/email tool calls, retries, token usage, latency, and per-metric feedback. The tracing helper is <code>src/story_tracing.py</code> .
Baseline run	Baseline run <code>data/eval/runs/day3_eval_results_20260624021839.json</code> : 43 cases, 6 passed, 37 failed, strict pass rate 14.0%, average score 0.861, p95 latency 5266.71 ms, average estimated cost \$0.000416, total estimated cost \$0.017869. LangSmith project: <code>kids-daily-story-agent</code> ; dataset id: <code>d5ea1870-1010-4fce-a880-f2b6565a6e80</code> .
Failure analysis	Top failures were <code>moral_alignment</code> in 27 cases, <code>required_interests_included</code> in 20 cases, and <code>age_appropriate_length</code> in 7 cases. Secondary failures were child safety false/real term hits in 6 cases, theme alignment in 5 cases, and repetition in 2 cases.
Improvement hypotheses	Prompt/control-flow: make the theme explicit and ask for parent-note moral wording, predicted lift on <code>moral_alignment</code> and <code>theme_alignment</code> . Prompt: list every required interest and require checker-visible wording, predicted lift on <code>required_interests_included</code> . Prompt/safety: pass all profile and global avoided terms directly into generation, predicted lift on <code>child_safe_content</code> . Length control: compute age-specific word budget before generation, predicted lift on <code>age_appropriate_length</code> .

Field	Completed Entry
Post-improvement run	Implemented the prompt changes in <code>src/story_agent.py</code> and verified 33 local tests plus dataset validation. A full live post-improvement LangSmith run was attempted on June 24, 2026, but the provider/LangSmith process hung without producing a result file and was stopped; measured post-run delta is therefore pending.
What is next	Re-run <code>scripts/day3_run_langsmith_eval.py --skip-langsmith-dataset-sync</code> with unbuffered output or provider timeout settings, then compare against <code>day3_eval_results_20260624021839.json</code> . If another week were available, add an LLM-as-judge moral/faithfulness evaluator calibrated against 10 human-reviewed cases and production monitors for safety failure rate, strict pass rate, p95 latency, cost, and tool failure rate.

Baseline Failure Clusters

Failure Mode	Count	Example Case	Likely Root Cause	Rough Failed-Case Cost
moral_alignment	27	happy_age4_blocks_patience	The prompt asked for a gentle lesson but did not require evaluator-visible moral language.	Included in \$0.015371 failed-case estimated total.
required_interests_included	20	happy_age8_forest_confidence	The model often used one interest naturally but dropped another concrete required interest.	Included in \$0.015371 failed-case estimated total.
age_appropriate_length	7	known_failure_exact_character_name_leak	The prompt used broad word-count guidance rather than an age-specific budget computed before generation.	Included in \$0.015371 failed-case estimated total.

Implemented Improvement

The generator now computes an age-specific word budget, builds explicit interest and avoided-topic lists, fixes the Montessori typo, requires the selected theme to appear in generated fields, asks for the lesson in `parent_note`, and tells the model to avoid all profile and global sensitive terms. This is intentionally a narrow prompt/control-flow change so the evaluation harness remains the source of truth.

Verification

```
.\.venv\Scripts\python.exe -m pytest tests\test_story_eval.py tests\test_story_agent.py
33 passed

.\.venv\Scripts\python.exe scripts\validate_day1_dataset.py
43 golden cases, valid: true
```

Production Monitoring Plan

Track strict case pass rate and child-safety failures on a rolling sample of production traces. Alert if p95 latency exceeds 15 seconds, estimated story-generation cost exceeds \$0.05, child-safety failures exceed zero in reviewed traces, required-interest coverage drops below 90%, or any optional tool failure rate exceeds 5% over one hour.